

ORBIT Fixed-EH $d \leq 6$ End-to-End Match Derivation

Purpose

This note gives a single reviewable argument for the strongest fixed-EH claim currently validated in ORBIT:

- after applying the specific witness solution found by the comparison pipeline, and
- after applying the specific nontrivial higher-derivative field redefinition,

the EFT truncated through ORBIT mass dimension 6 matches the fixed Einstein-Hilbert reference.

This note is meant to be read together with:

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artifacts/orbit_fixed_eh/orbit_fixed_eh_solution_case_validation_output.wl

- tests/artifacts/orbit_solution_case_manual_check_output.wl

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artifacts/orbit_fixed_eh/orbit_fixed_eh_quadratic_completion_analysis.wl

The goal here is not to restate every artifact mechanically. The goal is to show the full logic in one place and to manually verify the nontrivial step.

Claim

Let

$$L := \text{Log}[\bar{\mu}^2 / M^2].$$

On the positive branch used in the validation artifacts,

$$L = -3/2,$$

$$\text{LambdaRefFinal} = 0,$$

$$\text{KappaRefFinal} = 4 \sqrt{3} / M.$$

With those substitutions, the ORBIT-fixed-EH comparison through $d \leq 6$ has:

- exact agreement in every supported sector after reduction modulo IBP and field redefinitions
- one remaining nontrivial reduced discrepancy in sector $\{2,4\}$

and that remaining $\{2,4\}$ discrepancy is exactly reproduced by the quadratic Fierz-Pauli field-redefinition image

$$\begin{aligned} \delta h_{\mu\nu} &= -\frac{3}{20 M^2} \Box h_{\mu\nu} \\ &\quad -\frac{3}{10 M^2} \eta_{\mu\nu} \partial_\alpha \partial_\beta h^{\alpha\beta} \\ &\quad +\frac{3}{10 M^2} \eta_{\mu\nu} \Box h^\alpha{}_\alpha. \end{aligned}$$

Therefore the EFT and GR agree through ORBIT mass dimension 6 in the reduced/IBP sense used by the comparison pipeline.

1. Witness Solution From The Match Equations

The $d \leq 5$ exact-match artifact records the following three equations among the solved system:

$$\begin{aligned} \kappa \sqrt{-1 / (M^2 \kappa^2 (1 + L))} (3 + 2 L) &= 0 \\ \text{KappaRefFinal} &= 2 \sqrt{6} \kappa \sqrt{-1 / (M^2 \kappa^2 (1 + L))} \\ \sqrt{6} M^4 \kappa \sqrt{-1 / (M^2 \kappa^2 (1 + L))} (3 + 2 L) \\ &= 4 \text{KappaRefFinal} \text{LambdaRefFinal} \end{aligned}$$

Under the validation assumptions

$$M > 0, \kappa > 0, \bar{\mu} > 0,$$

the square-root prefactor is nonzero, so the first equation gives immediately

$$3 + 2 L = 0$$

and hence

$$L = -3/2.$$

Substituting $L = -3/2$ into the second equation:

$$1 + L = -1/2$$

so

$$\begin{aligned} & \sqrt{-1 / (M^2 \kappa^2 (1 + L))} \\ &= \sqrt{2 / (M^2 \kappa^2)} \\ &= \sqrt{2} / (M \kappa). \end{aligned}$$

Therefore

$$\begin{aligned} \text{KappaRefFinal} \\ &= 2 \sqrt{6} \kappa * \sqrt{2} / (M \kappa) \\ &= 4 \sqrt{3} / M. \end{aligned}$$

Finally, the third equation becomes

$$0 = 4 \text{KappaRefFinal} \text{LambdaRefFinal}.$$

Since $\text{KappaRefFinal} = 4 \sqrt{3} / M$ is nonzero, we get
 $\text{LambdaRefFinal} = 0$.

So the specific witness solution used throughout the validation is not mysterious:

$$\begin{aligned} \text{Log}[\bar{\mu}^2 / M^2] &= -3/2, \\ \text{LambdaRefFinal} &= 0, \\ \text{KappaRefFinal} &= 4 \sqrt{3} / M. \end{aligned}$$

2. What Already Matches Before Any Nontrivial Field Redefinition

After applying that witness solution, the manual-check artifact records

$$\text{D6}["\text{SupportedInteractionFieldRedefinitions}"] = \langle |\{3,2\} \rightarrow 0, \{4,2\} \rightarrow 0| \rangle$$

and

$$\text{D6}["\text{SupportedSectorResidualExpressions}"] =$$

$$\langle | \{1,0\} \rightarrow 0, \{2,0\} \rightarrow 0, \{2,2\} \rightarrow 0, \{3,0\} \rightarrow 0, \{3,2\} \rightarrow 0, \\ \{4,0\} \rightarrow 0, \{4,2\} \rightarrow 0, \{5,0\} \rightarrow 0, \{6,0\} \rightarrow 0 | \rangle.$$

This means:

- the cosmological sectors match
- the quadratic two-derivative sector matches
- the cubic and quartic two-derivative GR sectors match
- the quintic and sextic zero-derivative sectors match

So the only remaining nonzero piece at $d \leq 6$ is the unsupported quadratic four-derivative sector $\{2,4\}$.

That is why the proof can focus entirely on that sector.

3. The Remaining $\{2,4\}$ Difference

The validated projected $\{2,4\}$ discrepancy can be written as

$$\begin{aligned} \Delta L_{\{2,4\}} &= \frac{9}{10} M^2 B1 \\ &\quad - \frac{9}{20} M^2 B2 \\ &\quad - \frac{3}{5} M^2 B3 \\ &\quad + \frac{3}{10} M^2 B4 \\ &\quad - \frac{3}{20} M^2 B5, \end{aligned}$$

where the five projected basis monomials are

$$B1 := (\partial_b \partial_c h^{\alpha}_{\alpha}) (\partial_d \partial^d h^{bc})$$

$$B2 := (\partial_b \partial^b h^{\alpha}_{\alpha}) (\partial_d \partial^d h^{\beta}_{\beta})$$

$$B3 := (\partial_a \partial_b h_{cd}) (\partial^c \partial^d h^{ab})$$

$$B4 := (\partial_b \partial_d h_{ac}) (\partial^c \partial^d h^{ab})$$

$$B5 := (\partial_c \partial_d h_{ab}) (\partial^c \partial^d h^{ab}).$$

In coordinate form,

$$\Delta = (9/10, -9/20, -3/5, 3/10, -3/20) / M^2.$$

This is exactly the coordinate vector stored in the quadratic-completion artifact.

4. The Nontrivial Field Redefinition

The only nonzero validated field redefinition is

$$\begin{aligned} \delta h_{\mu\nu} &= -\frac{3}{10} M^2 \eta_{\mu\nu} \partial_\alpha \partial_\beta h^{\alpha\beta} \\ &\quad -\frac{3}{20} M^2 \Box h_{\mu\nu} \\ &\quad +\frac{3}{10} M^2 \eta_{\mu\nu} \Box h^{\alpha\beta}{}_{,\alpha\beta}. \end{aligned}$$

It is convenient to split this into three basis shifts:

$$\begin{aligned} \delta h^{\{1\}}_{\mu\nu} &:= \eta_{\mu\nu} \partial_\alpha \partial_\beta h^{\alpha\beta} \\ \delta h^{\{2\}}_{\mu\nu} &:= \Box h_{\mu\nu} \\ \delta h^{\{3\}}_{\mu\nu} &:= \eta_{\mu\nu} \Box h^{\alpha\beta}{}_{,\alpha\beta}. \end{aligned}$$

Then

$$\begin{aligned} \delta h_{\mu\nu} &= a_1 \delta h^{\{1\}}_{\mu\nu} \\ &\quad + a_2 \delta h^{\{2\}}_{\mu\nu} \\ &\quad + a_3 \delta h^{\{3\}}_{\mu\nu} \end{aligned}$$

with

$$\begin{aligned} a_1 &= -3 / (10 M^2), \\ a_2 &= -3 / (20 M^2), \\ a_3 &= 3 / (10 M^2). \end{aligned}$$

5. Why This Cannot Spoil The GR Cubic Or Quartic Match

This is the key EFT-counting step.

In ORBIT's sector language, the Einstein-Hilbert reference through mass dimension 6 contains the supported sectors

$\{1,0\}$, $\{2,0\}$, $\{2,2\}$, $\{3,0\}$, $\{3,2\}$, $\{4,0\}$, $\{4,2\}$, $\{5,0\}$, $\{6,0\}$.

The nontrivial shift $\delta h_{\mu\nu}$ contains one field and two derivatives, so it lives in sector $\{1,2\}$.

A first-order variation of an EH sector $\{n,2\}$ under $h \rightarrow h + \delta h$ therefore lands in $\{n,4\}$:

- $\{2,2\} \rightarrow \{2,4\}$
- $\{3,2\} \rightarrow \{3,4\}$
- $\{4,2\} \rightarrow \{4,4\}$

But ORBIT truncates by $nh + Nd \leq 6$, so:

- $\{2,4\}$ survives
- $\{3,4\}$ has total degree 7 and is truncated away
- $\{4,4\}$ has total degree 8 and is truncated away

This is the decisive reason the quadratic higher-derivative shift does not ruin the ordinary GR cubic or quartic sectors:

- the sectors $\{3,2\}$ and $\{4,2\}$ already matched exactly before this step
- the field redefinition can only modify $\{2,4\}$ within the $d \leq 6$ truncation

So the entire $d \leq 6$ problem really does reduce to the quadratic sector.

6. Manual Reconstruction Of The Quadratic Difference

The quadratic-completion artifact gives the projected image vectors of the three independent shifts above:

$$v_1 = (-2, 0, 2, 0, 0)$$

$$v_2 = (2, -1, 0, -2, 1)$$

$$v_3 = (2, -2, 0, 0, 0).$$

These are the $\{B_1, B_2, B_3, B_4, B_5\}$ coordinates of the first-order Fierz-Pauli shift produced by

$$- \delta h^{(1)}$$

$$- \delta h^{(2)}$$

$$- \delta h^{(3)}$$

respectively.

To prove that the remaining discrepancy is pure field-redefinition data, we solve by hand

$$a_1 v_1 + a_2 v_2 + a_3 v_3 = \Delta.$$

Using the third component:

$$2 a_1 = -3 / (5 M^2)$$

so

$$a_1 = -3 / (10 M^2).$$

Using the fourth component:

$$-2 a_2 = 3 / (10 M^2)$$

so

$$a_2 = -3 / (20 M^2).$$

Using the second component:

$$-a_2 - 2 a_3 = -9 / (20 M^2).$$

Substituting $a_2 = -3 / (20 M^2)$ gives

$$3 / (20 M^2) - 2 a_3 = -9 / (20 M^2),$$

hence

$$a_3 = 3 / (10 M^2).$$

Now check the remaining components:

First component:

$$\begin{aligned} & -2 a_1 + 2 a_2 + 2 a_3 \\ &= 6/(10 M^2) - 3/(10 M^2) + 6/(10 M^2) \\ &= 9/(10 M^2). \end{aligned}$$

Fifth component:

$$a_2 = -3 / (20 M^2).$$

So all five components match exactly:

$$\begin{aligned} & a_1 v_1 + a_2 v_2 + a_3 v_3 \\ &= (9/10, -9/20, -3/5, 3/10, -3/20) / M^2 \\ &= \Delta. \end{aligned}$$

This is the manual confirmation of the nontrivial step:

- the leftover $\{2,4\}$ discrepancy lies exactly in the image of a higher-derivative field redefinition
- the coefficients are exactly the ones recorded by ORBIT

No curvature-squared term is needed for this reconstruction.

7. Correct End-To-End Conclusion

Let

$$L_{\text{EFT}}^{\leq 6}$$

denote the ORBIT EFT truncated to sectors with $nh + Nd \leq 6$,
and let

$$L_{\text{GR}}^{\leq 6}$$

denote the fixed Einstein-Hilbert reference truncated the same way.

After inserting

$$\text{Log}[\bar{\mu}^2 / M^2] = -3/2,$$

$$\text{LambdaRefFinal} = 0,$$

$$\text{KappaRefFinal} = 4 \sqrt{3} / M,$$

the exact artifacts show that all supported reduced sectors match, and that the only remaining reduced discrepancy is the quadratic four-derivative sector:

$$L_{\text{EFT}}^{\leq 6} - L_{\text{GR}}^{\leq 6} = \Delta L_{\{2,4\}}.$$

The manual computation above shows

$$\Delta L_{\{2,4\}}$$

$$= \delta L_{\text{FP}}[\delta h]$$

for

$$\delta h_{\{\mu\nu\}}$$

$$= -\frac{3}{20 M^2} \Box h_{\{\mu\nu\}}$$

$$-\frac{3}{10 M^2} \eta_{\{\mu\nu\}} \partial_{\alpha} \partial_{\beta} h^{\{\alpha\beta\}}$$

$$+\frac{3}{10 M^2} \eta_{\{\mu\nu\}} \Box h^{\alpha\beta} \partial_{\alpha} \partial_{\beta}.$$

Because this shift only affects sector $\{2,4\}$ within the $d \leq 6$ truncation, the correct comparison statement is:

the original EFT and the fixed-EH reference agree on every supported reduced sector,

and the remaining reduced $\{2,4\}$ discrepancy is exactly the projected Fierz-Pauli image of δh .

This is the precise sense in which the fixed-EH EFT matches GR through mass dimension `6`.

8. Important Correction From The Step-0 Audit

The stronger literal statement

$$L_{\text{EFT}}^{\leq 6}(h) = L_{\text{GR}}^{\leq 6}(h + \delta h)$$

is **not** correct if $L_{\text{GR}}^{\leq 6}$ is taken to mean the raw unreduced ORBIT reference representative produced by `ExpandFixedEHReferenceLagrangian`.

The fresh step-0 validator starting from the original `.mx`` input shows:

- the raw full-reference first-order shift contains sectors `{1,4}` and `{2,4}`
- the extra `{1,4}` contribution reduces to zero
- the projected `{2,4}` coordinates of that raw full-reference shift are actually `{0, 0, 0, 0, 0, 0}`

So the naive substitution into the full unreduced ORBIT reference does **not** directly produce the validated quadratic discrepancy.

What *is* correct, and what ORBIT actually proves, is the reduced/IBP statement:

- the supported sectors of the original EFT match the witness-fixed GR reference
- the remaining reduced `{2,4}` discrepancy matches the projected Fierz-Pauli image of δh

That is the correct end-to-end formulation.

9. What Was Checked Manually Here

Manual checks performed in this note:

- solving the witness branch from the stored match equations
- proving by sector counting that the nontrivial field redefinition cannot alter $\{3,2\}$ or $\{4,2\}$ within the $d \leq 6$ truncation
- solving the quadratic image-decomposition problem by hand and recovering the exact coefficients of δh

Artifact-backed facts used here:

- the supported-sector residuals are all zero after the witness substitution
- the projected $\{2,4\}$ difference coordinates are the five numbers shown above
- the three independent quadratic field-redefinition image vectors are exactly v_1 , v_2 , v_3

So this note is not a replacement for the exact Mathematica artifacts. It is the analytic bridge that explains them end to end.